

IBM Db2 Overview, Architecture & Use Cases DBMS Module 6

> **Definition:** IBM Db2 is an enterprise-grade Relational Database Management System (RDBMS) developed by IBM, engineered for high-performance transactional processing (OLTP), analytical workloads (OLAP), and hybrid data management (HTAP) across hybrid cloud environments.

1. Detailed Technical Explanation

1. Key Features of IBM Db2:

- **BLU Acceleration:** In-memory columnar processing for multi-fold speedup in analytical queries.
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2. IBM Db2 Product Editions

Db2 Edition	Target Environment	Hardware / Resource Limits	Key Features
Enterprise Edition	Large enterprises, mission-critical applications	Unlimited	Full functionality, including HTAP, advanced security, and scalability
Express Edition	Small to medium businesses, development and testing	Up to 16GB memory, 10GB storage	Core DBMS functionality, simplified deployment
Express-CE Edition	Cloud environments, small businesses	Up to 16GB memory, 10GB storage	Cloud-optimized, simplified deployment

3. Industry Use Cases

- Banking & Financial Services:** Core banking transactions, real-time credit card fraud detection using pureScale high availability.
 - Healthcare:** Patient data management, clinical research analytics.
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4. Comparison: IBM Db2 vs Other RDBMS

Feature	IBM Db2	Oracle Database	MySQL / PostgreSQL
Performance	High (BLU Acceleration)	High	Medium
Scalability	High (Scale-Out)	Medium	Medium
Security	Advanced (Security Guard)	Advanced	Basic
Cloud Support	Strong (Cloud Pak for Data)	Medium	Strong

5. Core Concepts & Memory Keywords

- **pureScale:** IBM active-active database clustering for zero downtime.

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6. Must-Write Points for Exams

- IBM Db2 supports Hybrid Transactional and Analytical Processing (HTAP) in a single engine.

- Db2

7. Quick Recall Flow

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